

BT401 final solved paper:

Write two derived resources 2

2. Derived Genetic Resources

- Obsolete varieties
- Breeding lines with particular genes and performances
- Advanced cultivars
- Parents of hybrid varieties
- Cytogenetic stocks/ tester
- Mutants

What are the needs of genetic preservation 2

A **genetic preservation** is the first step in the cloning process, allowing you to produce an identical **genetic** twin or clone, which will be born at a later place in time.

A genetic preservation serves as an insurance policy for breeders and owners of valuable cattle by enabling them to extend and develop a specific bloodline when additional production is needed or untimely losses or reproductive inabilities occur.

The DNA, cryogenically preserved from the tissue sample, can permanently store the genotype of the elite donor animal, providing a genetic blueprint to recreate that cow, bull, heifer - or even a steer - at any time in the future.

Define tertiary gene pool 3

Tertiary gene pool

- Members of this gene pool are more distantly related to the members of the primary gene pool. The primary and tertiary gene pools can be intermated, but gene transfer between them is impossible without the use of "rather extreme or radical measures" such as:
 - Embryo rescue (or embryo culture, a form of plant organ culture)
 - Induced polyploidy (chromosome doubling)
 - Bridging crosses (e.g., with members of the secondary gene pool)

What is pakistan animal resource management program 3

Pakistan is endowed with diverse **livestock genetic resources**. In fact, it is postulated that one of the centers of **animal** domestication lies in this part of the world. **Pakistan** has nearly 50 million goats. Goats are kept for milk and meat production and contribute significantly to the income of the rural farmers.

Snow Leopard	
•	Alpine Markhors

•	Asiatic Cobra
•	Mugger Crocodile
•	Himalayan Brown Bear
•	Indus River Dolphin
•	Asian Black Bear
•	Fishing Cat

What is low oxygen and low pressure storage 3

Low-pressure and Low-oxygen Storage:

For conservation of cultured plant materials low-pressure storage (LPS) and low-oxygen storage (LOS) have been developed. These are alternative methods of cryopreservation and cold storage.

What are the advantages of germplasm storage 5

Storage of seed is indispensable to most of plantation forestry, and the practice should not be dismissed too readily as a basic tool in maintaining genetic diversity. Conventional seed storage offers several advantages:

1. Seeds of many valuable species can survive long-term storage ('longterm' is defined as spanning a period of time longer than one rotation)
2. Good storage facilities are now available in most of the world, and they are used extensively for tree seed storage for various regeneration purposes;
3. Seed storage is a relatively cheap method for conserving a broad range of germplasm
4. Large land areas are not tied up in conservation
5. International exchange of genetic material is facilitated by seed storage.

What are the approaches for preservation 5

The debates regarding the best methods of digital preservation have been underway for many years (Verdegem 2006: 8). Various theoretical solutions have been proposed but each has serious flaws, such as:

- Setting of standards;
- Preserving the original technology used to create or store the records;

Migrating the software necessary to retrieve, deliver and use the records;

- Preserving records in governmental bodies;
- Conversion; and
- Emulating the original technology on new platforms.

What are the objective of research institutes for livestock

Major development objectives

The primary objectives^s of the Roundtable were to:

- provide a forum for senior livestock scientists and developers for the exchange of views and experiences; and
- raise the level of awareness of a far wider and influential audience with regard to the potential and the constraints facing animal agriculture in low income countries.

⁵ This sections draws on the initial presentation of Simeon Ehui, who was the principal organizer of the Roundtable at ILRI

Specific immediate objectives

The immediate objectives of the Roundtable were to:

- review the contribution and potential of livestock to increase sustainable food production, and contribute to income generation in low income countries with a forward perspective to 2020 ("The Global 2020 Vision for Livestock");
- identify major social, economic, technical and institutional constraints limiting livestock's contribution to achieving food security and economic development; and
- define appropriate strategies to alleviate these constraints and propose a framework for international action to enhance animal productivity in its broadest sense.

Expected outputs

The expected outputs of the Roundtable were:

- an analysis of past and present trends in livestock productivity and consumption of livestock products which would be used in part as an input to a "2020 Vision" paper to be further developed after the meeting;
- a statement of a defined set of objectives within the time frame specified and a related description of the constraints that must be overcome for the objectives to be achieved; and
- formulation of the set of measures ("a framework for action") or strategies needed for increasing livestock productivity in low income countries and securing better management of the natural resource base from the present to the end of the second decade of the 21st century.

Mcqs easy but from the entire course

BT401

What is phenylketonuria? 2

Phenylketonuria (PKU)

It is a human genetic condition caused by mutations to a gene coding for a particular liver enzyme. In the absence of this enzyme, an amino acid known as phenylalanine does not get converted into the next amino acid in a biochemical pathway, and therefore too much phenylalanine passes into the blood and other tissues. Change in environment (lowering Phenylalanine consumption) can affect the phenotype of a particular trait, demonstrating a gene-environment interaction

Define threatened species? 2

Threatened species are any species (including animals, plants, fungi, etc.) which are vulnerable to endangerment in the near future. Species that are threatened are sometimes characterized by the population dynamics measure of critical dispensation, a mathematical measure of biomass related to population growth rate

State 2 quarantine principle? 2

Isolation and **quarantine** help protect the public by preventing exposure to people who have or may have a contagious disease. ... **Quarantine** separates and restricts the movement of people who were exposed to a contagious disease to see if they become sick

Modern quarantine does not have to be absolute to be effective

- Even partial or “leaky” quarantine can reduce disease spread
- Partial quarantine can be an effective supplement to vaccination

Explain habitat loss cause of extinction? 3

Habitat destruction is the process in which natural **habitat** is rendered unable to support the **species** present. In this process, the organisms that previously used the site are displaced or destroyed, reducing **biodiversity**. ... **Habitat destruction** is currently ranked as the primary **cause** of **species extinction** worldwide.

What are molecular marker? 3

In genetics, a **molecular marker** (identified as **genetic marker**) is a fragment of DNA that is associated with a certain location within the genome. **Molecular markers** are used in **molecular** biology and biotechnology to identify a particular sequence of DNA in a pool of unknown DNA.

- RFLP (or Restriction fragment length polymorphism)
- SSLP (or Simple sequence length polymorphism)
- AFLP (or Amplified fragment length polymorphism)
- RAPD (or Random amplification of polymorphic DNA)
- VNTR (or Variable number tandem repeat)

What are three common function of quarantine? 3

Quarantine practices in most countries have at least three common functions.

- a. The first is exclusion or regulatory actions to prevent or reduce the risk of entry of exotic pathogens, pests, or parasites along artificial pathways.
- b. Second is the containment, suppression, or eradication of pests or pathogens that have been recently introduced.
- c. Third is the assisting of exporters to meet the quarantine requirements of importing countries.

What are mechanism of cryopreservation name steps? 5

Until two decades ago the genetic resources were getting depleted owing to the continuous depredation by man. It was imperative therefore that many of the elite, economically important and endangered species are preserved to make them available when needed. Many methodologies have been devised for long term preservation of material.

Cryo-preservation

Cryo is Greek word. (krayos – frost). It literally means preservation in “frozen state.”

Cryo-preservation or **cryo-conservation** is a process where organelles, cells, tissues, extracellular matrix, organs or any other biological constructs susceptible to damage caused by unregulated chemical kinetics are preserved by cooling to very low temperatures (typically -80 °C using solid carbon dioxide or -196 °C using liquid nitrogen)

Cryopreservation can be done at

•	Over solid carbon dioxide (at -79 degree)
•	Low temperature deep freezer (at -80 degree)
•	In vapor phase nitrogen (at -150 degree)
•	In liquid nitrogen (at -196 degree)

60: Cryopreservation Technique:

A **cryoprotectant** is a substance used to protect biological tissue from freezing damage. Due to ice crystal formation cell membrane and cell integrity can ruptured. **cryoprotectants** are usually antifreeze compound.

Types of Cryoprotectants:

1. **Membrane permitting** which can freely diffuse the membrane such as **glycerol (G)**, **ethylene glycol (EG)** and **dimethyl sulfoxide (DMSO)**
2. **Non Membrane permitting** which cannot **permeate** the cell membrane such as **sugars**

Cryopreservation technique:

•	Slow freezing and thawing
•	Rapid freezing and thawing

- | |
|---------------|
| Vitrification |
|---------------|

Ultrarapid freezing

DNA Bank. 5

DNA banks can now be considered as a means of complimentary conservation. DNA storage is particularly useful for those species that cannot be conserved in traditional seed or field genebanks and nor conserved in situ due to high risk in that area.

Advantages

DNA banking is an efficient, simple and long-term method to conserve the genetic information.

Disadvantages

There are problems with subsequent gene isolation, cloning and transfer of DNA back to a plant and it currently does not allow the regeneration of the same genotype as the original sample.

Storage strategy

Determining what to store and for how long is an important consideration, used to determine sample size, capacity of the DNA bank, preparation of samples and documentation. Long-term needs and expected volume and number of samples to be stored will determine organization and repository design.

Processing of samples

DNA preserved in DNA banks will be stored either within cells and extracted upon retrieval from storage or extracted from cells and purified before storage. The quality of the DNA is expressed through yield, purity, molecular weight, amplification efficiency and authenticity of sequences. The quality of DNA extracted from plant specimens is dependent on the condition of the specimen before storage, the storage environment and the duration of storage. Rapid drying of plant samples with silica gel or lyophilisation helps to preserve the DNA.

S torage

Once extracted DNA is a stable biomolecule, although it can easily be degraded during extraction and storage. Quality declines within days in hydrated samples held at room temperature or in refrigerators. Drying the sample or storing it in freezers or liquid nitrogen achieves better preservation of DNA molecular size. For this reason, DNA is better conserved in a form that is close to the original state and most DNA banks store cells or tissues and extract DNA upon request.

There is little information on the long-term stability of extracted DNA during frozen storage, but most repositories consider several years to decades as realistic. Information on the stability of purified DNA dissolved in buffer suggests that the overall fragment size decreases with storage time, and that the usefulness of the specimen for PCR-based assays may be 1–2 years when stored at 4 °C, 4–7 years when stored at -18 °C and greater than 4 years when stored at -80 °C (Madisen et al. 1987; Visvikis et al. 1998).

The choice of temperature usually depends on the moisture level within the sample

Difference b/w in breeding and out breeding. 5

Inbreeding:

“**Inbreeding**, the mating of individuals or organisms that are closely related through common ancestry.”

Outbreeding:

“The intentional breeding of distantly related or unrelated individuals for the purpose of producing offspring”

ng of superior quality.

Biological diversity. 2

Biodiversity, a portmanteau of biological (life) and diversity, generally refers to the variety and variability of life on Earth. According to the United Nations Environment Programme (UNEP), **biodiversity** typically measures variation at the genetic, species, and ecosystem level.

Questions

1. Define extinction?

“A species becomes extinct when the last existing member of that species dies.”

Extinction, in biology the dying out or termination of a species.

Extinction occurs when species are diminished because of ;

•	Environmental forces (habitat fragmentation, global change, overexploitation of species for human use)
•	Evolutionary changes in their members (genetic inbreeding, poor reproduction, decline in population numbers).
•	

Extinction becomes a certainty when there are no surviving individuals that are able to reproduce and create a new generation. A species may become functionally extinct when only a handful of individuals survive, which are unable to reproduce due to;

Poor health,	•
Age,	•
Sparse distribution over a large range,	•
A lack of individuals of both sexes (in sexually reproducing species)	•

3. Four main mechanism of evolution?

The allele frequency in a population's gene pool can be affected by evolutionary mechanisms, such as;

•	Mutation
•	Gene flow
•	Inbreeding
•	Natural selection,
•	Founder effect
•	Random genetic drift.

BT401

Germplasm Conservation 10

In 1972, conservation of habitats rich in genetic diversity was recommended in the UN conference. Then an International Board for Plant Genetic Resource (IBPGR) was established. This board has objectives to provide necessary support for collection, conservation and utilization of plant genetic resources from anywhere in the world.

(a) In situ Conservation:

Since 1980, in situ conservation has received high priority in the world conservation strategy. The method of conservation is to preserve land races with wild relatives in which genetic diversity exists.

(b) Ex situ Conservation

It is the chief mode of conservation of genetic resources including both cultivated and wild ones. Under suitable conditions genetic resources are conserved for a long term as gene bank. Such gene bank is of two types:

- In vivo Gene Bank
- In vitro Gene Bank

(i) In vivo Gene Bank:

Generally plant seeds, vegetative propagules are used for storage for long time. The whole plants are preserved. This type of conservation strategy is called in vivo gene bank. In this approach, conservation method of storage is used for preservation of plant genetic resources

(ii) In vitro Gene Bank:

3. This approach includes the conservation of genetic resources by non-conventional methods. In this approach explants are grown on medium.

Methods of Preservation

Free Preservation or Cryopreservation:

Cryopreservation (Latin Kuos means frost) means storage of materials at very low temperature. Plant cells and tissue cultures are brought to zero state of metabolism by subjecting them to ultra-low temperature i.e. -196°C . It is done by using liquid nitrogen which provides approximately -496°C . Cryoprotectants (e.g. glycerol, proline, mannitol, dimethylsulfoxide, sorbitol) are also used to protect the viable cells from the damage during freezing and thawing (to become unfrozen or warm). Germplasm of some plants (in the form of shoot tips, nodal or meristem explant culture) are stored at low and nonfreezing temperature ($1-9^{\circ}\text{C}$). At low temperature, growth of plant material is slow down but not completely stopped as in cryo-preservation. In cold storage there is no risk of cold injuries.

Low-pressure and Low-oxygen Storage:

For conservation of cultured plant materials low-pressure storage (LPS) and low-oxygen storage (LOS) have been developed. These are alternative methods of cryopreservation and cold storage.

Germplasm Exploration Laboratory 5

Germplasm Exploration Laboratory:

Plant exploration is the avenue to germplasm for crop improvement, which cannot be obtained by exchange. The spread of improved varieties has resulted in the loss of indigenous crop genetic

diversity. Whereas, industrialization and urbanization has seriously damaged the natural environment of crop plants. Human population is rapidly eroding the reservoir of genetic diversity. Habitats are disappearing at an alarming rate as forests are cleared, roads and cities expand, grasslands are plowed, burned and overgrazed, land is inundated by lakes impounded by new dams and new lands are irrigated for increased production of modern cultivars. The face of the earth is changing at an accelerating and alarming rate; and as it changes, more genetic diversity is lost forever. The Plant Exploration Laboratory has organized more than hundred expeditions in different agro-ecological regions of Pakistan to collect the targeted plant species. The main emphasis is to collect the major crops and their wild relatives as these species are under threat

Gene flow 5

Gene Flow:

“The introduction of genetic material (by interbreeding) from one population of a species to another”

Vertical Gene transfer: “The transfer of genes from parents to offspring.”

Horizontal gene transfer:

“Horizontal gene transfer is known to occur between different species, such as between prokaryotes and eukaryotes, between the three DNA-containing organelles of eukaryotes, the nucleus, the mitochondrion and the chloroplast.

Steps for conservation of plant genetic resources 5

The Global Strategy for Plant Conservation

The **Global Strategy for Plant Conservation (GSPC)** is a program of the UN's Convention on

Biological Diversity founded in 1999. It is a Plan to Save the World's Plant Species - grew out of the Convention on Biological Diversity and is being fed into government policy around the world.

Vision of GSPC:-

“Without plants, there is no life. The functioning of the planet, and our survival, depends on plants. The Strategy seeks to halt the continuing loss of plant diversity”

Objectives:

The GSPC highlights the importance of plants and the ecosystem services they provide for all life on earth, and aims to ensure their conservation. The GSPC has 5 main objectives:

1. Plant diversity is well understood, documented and recognized
2. Plant diversity is urgently and effectively conserved
3. Plant diversity is used in a sustainable and equitable manner
4. Plant diversity is used in a sustainable and equitable manner
5. The capacities and public engagement necessary to implement the strategy have been developed.

Critical Endangered Species.

A **critically endangered** (CR) species is one which has been categorized by the International Union for Conservation of Nature (IUCN) as facing an extremely high risk of extinction in the wild.

Examples:

Eastern Lowland Gorilla

Hawksbill Turtle

Javan Rhino

Malayan Tiger

MC160400524: Saba Iqbal

BT401:Quiz #2

Question # 6 of 10 (Start time: 05:52:44 AM, 27 August 2018)

Which species has been reintroduced in Lal Suhanra National Park

Select the correct option



wild ass



leopard



black buck



Neilgai

Question # 10 of 10 (Start time: 05:54:11 AM, 27 August 2018)

Main objective of Habitat-Species Management Area is

Select the correct option

☐	to maintain species
☐	to conserve species
☐	to conserve and maintain habitat
☒	all of these

Question # 9 of 10 (Start time: 05:53:46 AM, 27 August 2018)

which Category areas are not necessarily associated with human presence and intervention?

Rectar

Select the correct option

☒	Category I
☐	Category IV
☐	CategoryV
☐	Category VI

Question # 7 of 10 (Start time: 05:53:06 AM, 27 August 2018)

The main objective of category V is

Select the correct option

☐	to protect species
☐	to protect habitat
☐	to protect environment
☒	maintain a balanced interaction of nature and culture

MC160400524: Saba Iqbal

BT401:Quiz #2

Question # 2 of 10 (**Start time: 05:50:28 AM, 27 August 2018**)

Sir Sadiq Muhammad Khan Abbasi established

Select the correct option

☐	Lahore Safari Zoo
☒	Bahawalpur Zoo
☐	Hyderabad Zoo
☐	Karachi Municipal Aquarium

BT401:Quiz #2

Question # 10 of 10 (Start time: 07:46:30 PM, 27 August 2018)

Sir Sadiq Muhammad Khan Abbasi established

Select the correct option

☐	Lahore Safari Zoo
☒	Bahawalpur Zoo
☐	Hyderabad Zoo
☐	Karachi Municipal Aquarium

B1401:Quiz #2

Question # 9 of 10 (Start time: 07:46:21 PM, 27 August 2018)

Which species has been reintroduced in Lal Suhanra National Park

Select the correct option

☐	wild ass
☐	leopard
☒	black buck
☐	Neilgai

BT401:Quiz #2

Question # 8 of 10 (Start time: 07:45:42 PM, 27 August 2018)

The main objective of category V is

Select the correct option

☐	to protect species
☐	to protect habitat
☐	to protect environment
☒	maintain a balanced interaction of nature and culture

BT401:Quiz #2

Question # 7 of 10 (Start time: 07:45:10 PM, 27 August 2018)

Hingol National Park was declared reserved in

Select the correct option

☐	1990
☐	1989
☐	1991
☒	1988

B1401:Quiz #2

Question # 6 of 10 (Start time: 07:44:26 PM, 27 August 2018)

which Category areas are not necessarily associated with human presence and intervention?

Select the correct option

☒	Category I
☐	Category IV
☐	CategoryV
☐	Category VI

Question # 5 of 10 (Start time: 07:43:55 PM, 27 August 2018)

WCPA stands for

Select the correct option

☒	World Commission on Protected Areas
☐	World comity on Protected Areas
☐	World Conference on Protected Areas
☐	both a and c

BT401:Quiz #2

Question # 4 of 10 (**Start time: 07:43:01 PM, 27 August 2018**)

Which of the following category encourage the conservation of aquatic biodiversity?

Select the correct option

☐	Category III
☐	Category II
☒	Category V
☐	Category VI

BT401:Quiz #2

Question # 3 of 10 (Start time: 07:42:10 PM, 27 August 2018)

Main objective of Habitat-Species Management Area is

Select the correct option

☐	to maintain species
☐	to conserve species
☐	to conserve and maintain habitat
☒	all of these

BT401:Quiz #2

Question # 2 of 10 (Start time: 07:41:41 PM, 27 August 2018)

Which one of the following is the largest national park of Pakistan?

Select the correct option

☐	Chitral national park
☒	Hingol national park
☐	ayubia national park
☐	hazaraganji national park

BT401:Quiz #2

Question # 1 of 10 (Start time: 07:40:53 PM, 27 August 2018)

Ramsar convention was negotiated inby countries and NGOs

Select the correct option

☐	1980
☐	1990
☒	1960s
☐	1982

BT401:Quiz #1

Question # 1 of 10 (Start time: 08:32:56 PM, 06 June 2018)

Gene bank is an type of of

Select the correct option

☐	In Situ conservation
☐	Ex Situ conservation
☐	In vivo Conservation
☐	None of these

BT401:Quiz #1

Question # 2 of 10 (**Start time: 08:33:44 PM, 06 June 2018**)

areas of great genetic diversity are protected from human interference are known as

▶ **Select the correct option**

☐	Gene Bank
☐	Botanical garden
☒	Gene Sanctuary
☐	Zoological park

BT401:Quiz #1

Question # 3 of 10 (Start time: 08:35:03 PM, 06 June 2018)

animal genetic resources refereed as

Select the correct option

- | | |
|----------------------------------|---|
| ☐ | animal genetic resources for food and agriculture |
| ☐ | farm animal genetic resources |
| ☐ | livestock biodiversity |
| ☒ | All |

BT401:Quiz #1

Question # 4 of 10 (Start time: 08:35:25 PM, 06 June 2018)

PGRFA Stands for.....

Select the correct option

☐	Plant Genetic Resources for Agriculture
☒	Plant Genetic Resources for Food and Agriculture
☐	Plant Genetic resources And Forrest Agriculture
☐	Both A and B

BT401:Quiz #1

Question # 5 of 10 (Start time: 08:36:27 PM, 06 June 2018)

NCCP stands for.....

Select the correct option

☐	Natural cultural collection of Pakistan
☐	National cultural convention of Pakistan
☒	National Culture Collection of Pakistan
☐	none of these

BT401:Quiz #1

Question # 6 of 10 (Start time: 08:37:07 PM, 06 June 2018)

Conservation of plant genetic resources is necessary for

Select the correct option

☒	Food Security
☐	Agrobiodiversity
☐	Commercial use
☐	Both A and B

BT401:Quiz #1

Question # 7 of 10 (Start time: 08:38:22 PM, 06 June 2018)

Peste des petits ruminants also known as.....

▶ Select the correct option

☐	rinderpest,
☐	foot and mouth disease,
☒	sheep and goat plague.
☐	none of these

BT401:Quiz #1

Question # 8 of 10 (Start time: 08:39:16 PM, 06 June 2018)

According to FAO, Species of mollusks are approximately

Select the correct option

☐	31000
☒	85000
☐	47 000
☐	13 000

BT401:Quiz #1

Question # 9 of 10 (Start time: 08:39:54 PM, 06 June 2018)

GSPC was founded in

Select the correct option

☐	1999
☒	1990
☐	1989
☐	1992

BT401:Quiz #1

Question # 10 of 10 (Start time: 08:40:12 PM, 06 June 2018)

which disease was spread due to consuming mercury poisoned fishes by people ?

Select the correct option

☐	Parkinson Disease
☒	MINAMATA Disease
☐	Diarrhea
☐	Celiac Disease

BT401 final paper spring 2018

Total Q 50 (80 marks)

40 Mcqs 6 7 mcqs were from last quiz

6 short Q (2 and 3 marks)

1. What is botanical garden? (2)

2. Define forest genetic resources? (2)

3. On how many tinypoly tress rely on maintain the climate? (2) asy h kuch tha Q yad ni

4. Write the variations in variation gene? (3)

5. Define strict reserve conviction? (3)

6. Define seed storage? (3)

4 long Q (5 and 10 marks)

1. Write features of cartagona protocol? (5)

2. Write note on convention of botanical garden? (5)

3. What is cold storage. Discuss? (5)

4. Write the impact of climate change on FGR. Also discuss the climate change on organisms which associate with FGR? (10)

BT401 Final Term 8 AM 1st September ([Aleeze Tahir Shah](#))

1. Briefly explain state of world's forest genetic resources. 3
2. What are three pillars of Ramsar Convention? 5
3. What is the extreme weather effect on invertebrate's genetic resources? 5
4. Discuss the impact of climate change on FGR. Also discuss how this changing climate effect organism which is associated with trees? 10
5. What types of services are provided by PEPA? 5
6. What areas are covered by Cartagena protocol?
7. Briefly explain historical background of Ramsar convention. 3
8. Describe purpose of panel code. 2
9. What is location of cholistan wildlife sanctuary? 2
10. Mission of Ramsar convention.
11. MCQS were easy from mid and quiz

bt401

second largest national park?2

when and which city cartagena protocol?2

function of quarantine protocol?

distinguishing features of habitat species management area?

main objective of forest genetic resources?

objectives of cartagena protocol?

effect of climate change genetic resources of fgr?and the organisms associated with trees 10

BT401 Today Paper

40 mcq thy

Define Treaty (2)

Define Landsape and Seasacpe protected area (2)

What is cholistan protected area name (2)

Write the objectives of Mounment features (3)

Write the objective of socio asa hi tha kuch 5 marks ka

What is cold storage. Discuss? (5)

Write the impact of climate change on FGR. Also discuss the climate change on organisms which associate with FGR? (10)

Write the objectives of CBD (10)

Bt401

5 o clock

What is the mission of ramsar 2

What was the reason of botanical garden 2

Ak extinct specie ko protect krne wala Tha

Climate change effect of fda 10

Cold storage 10

Stratgies for Angr 5

al paper spring 2018

Total Q 50 (80 marks)

40 Mcqs 6 7 mcqs were from last quiz

6 short Q (2 and 3 marks)

MCqs:

1. Dolphin is aMammal.
2. Ramsar convention in.....1960.
3. 21% out of 5,488 mammal species and 12% out of 9,990 bird species are considered to be endangered.
4. WCPA stands forWorld commission on protected areas.
5. In Europe, one estimate put the value of "marketed non wood goods" from forests at €2.3 billion
6. Since the 1970s, the population of the Indus Dolphin has significantly increased here.
7. Animal genetic resources.....
8. Biggest national park of Pakistan.....
9. If hunting is prohibited, a game reserve may be considered a "Nature Reserve"
10. A sanctuary is a protected area which is reserved for the conservation of only animal and human activities like harvesting of timber
11. Bahawalpur Zoo covers.....25 acrs
12. Blackbuck became virtually extinct in the Cholistan Desert but the species has been re introduced in Lal Suhanra
13. Hingol National Park declared reserved in 1988
14. Natural geological and geomorphological features waterfalls

Baqi b easy thy

1. What is botanical garden? (2)
2. Define forest genetic resources? (2)
3. What is location of cholistan wildlife sanctuary? 2
- 4.Name of Botanical gardens in Pakistan.3
- 5.Strategies of Animal genetic resources 3
- 6.Plant genetic resources.3
- 7.Objectives of Cartagena Protocol on Biosafety 5
8. Write note on convention of botanical garden? (5)
- 9.What is the extreme weather effect on invertebrate's genetic resources? 5
10. Discuss the impact of climate change on FGR. Also discuss how this changing climate effect organism which is associated with trees? 10

MCqs:

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Baqi b easy thy

9/2, 10:30 AM] Azan: Paper:BT401

I). Who estblish bwp zoo?

II). Why does FGR matter?

III). Programmes of nation for conservation of PGR.

IV). Conservation status of hingol nationl prk.

V). In-vivo gene bank.

VI). Cold storage

VII). Socio cultlojopoml z hgycegz zklpural value of national monument feature.

VIII).

[9/2, 10:31 AM] Azan: Long ques wo tha 🤖🤖🤖🤖

Effect of climate change on FGR. And give effect of climate change on the tree associated organisms esabe tha kuch ..

paper bt 401

1.define zoological gardens (2)

2.different types of gene bank (2)

3.name of 3 botanical garden in pakistan (3)

4.state world forest genetic resource (3)

5.outcomes of introduction of exotic species in FGR (5)

6.socio cultural values of national monument features (5)

7.limitations of germplasm conservation (5)

8.services provided by PEPA (5)

9.Effect of Climatic changes on FGR.Also effect on organisms living on trees (10)

Bt401 today 2:30

1. Define phenotypic plasticity. 2

2. What are game species. 2

3. What are Habitat species management areas. 2

4. Areas covered by cartagena protocol. 3

5. History of Ramsar convention. 3
 6. Effect of extreme weather on invertebrate genetic resources. 5
 7. Socio cultural value of national monument features. 5
 8. Why do plant genetic resources matter?
 9. Impact of climate on FGR and animals associated to it. 10
- MCQs
1. FAO stands for?
 2. GEWIS stands for?
 3. Ramsar convention came into force in?
 5. Report on fish sperm cryopreservation was published in by Blaxter.
 6. Number of laboratories on plant genetic resources in Pakistan?
 7. Area covered by Bahawalpur Zoo?
 8. Genetic variations are the variations due to
Genes. alleles. Both a and b
 9. Reproductive isolation is
Sympatric. Allopatric
 10. Light intensity for most of the plants is s⁻¹ m⁻²
 11. Balck buck wala mcq tha
 12. Protected areas and ecological buffer type ka mcq.
Bass itna hi yaad hy.

BT401

Primary objectives of national park.
Germplasm conservation
Name atleast 3 botanical gardens
Phenotypic plasticity
Limitations of germplasm
Threats to FGR
Botanical gardens
Impact of climate change on FGR

BT401....

- 1.FGR... 2
- 2.effect of climate on FGR.. 2
3. History of ramsar convention... 3
- 4.areas covered by the Cartagena protocol.. 3
5. In vivo gene bank... 3
- 6.link b/w botanical garden and climate change.. 5
7. Name of labs hosted by botanical institute of genetic resources... 5
- 8.extreme weather effects on invertebrate's genetic resources... 5
9. Impact of climate change on FGR.... 10

Bt 401

What is the central goal of ant zoo?2
What type of material Dna bank is used?2

What is treaty?2
 Primary features of category VI?3
 Strategies of AnGR?3
 Variations of gene variations?3
 Cold storage?5
 Labs for plants genetic resources?5
 Three pillars of Ramsar?5
 Long same e tha

BT 401

- 1: What is primary objective National Park? 2M
- 2: What are game species? 2M
- 3: Define term phenotypic plasticity? 2M
- 4: Give 3 research Botanical garden? 3M
- 5: Write possible outcomes for introduction of exotic species in FGR? 3M
- 6: Give link between climate change and Botanical garden?3M
- 7: Explain method "Processing of sample" in DNA bank? 5M
- 8: How invertebrate genetic resources cope with climate change? 5M
- 9: Describe the features Pakistan Terrestrial Water and Maritime Zones Act 1976? 5M
- 10: Describe the impact of climate change on FGR? And also describe the effects of changing climate on organisms associated with trees? 10M

[#Quiz II](#) and [#FinalTerm](#) MCQ's

1. The main objective of Category V is..... maintain a balanced interaction of nature & culture.
2. Which Category area are not necessarily associated with presence and intervention..... Category-IV
3. Main objective of Habitat-Species Management Area is..... Option d
 a. to maintain species b. to conserve species c. to conserve & maintain habitat d. all of these
4. Sir Saqi Muhammad Khan Abassi established..... Bahawalpur Zoo
5. Which species has been reintroduced in Lal Suhanra National Park..... Black buck
7. Ramsar convention was negotiated_____by countries and NGO's.....1960
8. WCPA stands for.....World Commission on Protected Areas
9. Which of the following category encourage the conservation of aquatic biodiversity..... Category-V
10. Hingol National Park was declared reserved in.....1988
11. Which of the following is largest national park of Pakistan.....Hingol National Park
12. Large Natural areas comes into..... Category VI
13. Ramsar came into force in..... 1975
14. Total Ramsar sites are____in Pakistan.....19
15. Bahawalpur zoo cover..... 25 acres
16. In Game reserve the major focus is specially theanimals
17. Market value of non-wood goods from forest at..... €2.3 billion
18. Climate has direct effect on.....FGR
19. Trigger for migration..... all of these
 a. local climate b. local availability of food c. for mating reasons d. all of these

20. There are ____ type of Genome transfer.....2
21. ____ rupees will be fine for cutting tree in 1992 act.....5000
22. Cartagena Protocol on Bio-safety.....2001
23. Convention on biological diversity.....1992
24. Dolphin is a.....Mammal
25. 21% out of 5,488 mammal species & 12% out of 9,990 bird species are considered to be endangered.
26. Since the 1970's the population of Indus dolphin has significantly increased here.
27. Animal Genetic resources..... option d (all)
a. animal genetic resources for food & agriculture b. farm animal genetic resources c. livestock biodiversity d. all
28. If hunting is prohibited, a game reserve may be considered a Nature Reserve
29. National geological and geomorphological features..... waterfalls
30. Seaweeds population..... 13 000
31. Snow leopard population.....300
32. Land race also called.....weeds
33. ____are the link from generation to generation of all living matter..... Genes
34. PGRFA assets; facilitate to access to needed genetic materials and govern the sharing of the accruing benefits; add value to them through crop improvement; and sustainably intensify crop production as may be needed.
35. The 1st report on fish sperm cryopreservation was published by Blaxter..... (1953).
36. Vegetative propagation is alternative to.....Tissue culture
37. Using microarray analysis, detected that up to ____ genes were significantly induced by drought in two Mediterranean pine species.....113
38. FAO stand for..... Food & Agriculture Organization
39. GEWIS stands for..... Genome wide interaction scan
40. Number of laboratories on plant genetic resources in Pakistan.....6
41. Genetic variations are variations due to..... Genes
42. Reproductive isolation is.....Sympatric
43. Ensure a light intensity in the range from 10 to 1000 $\mu\text{mol S}^{-1} \text{m}^{-2}$.
44. Species extinct till yet..... 5 billion
45. Gene bank is a type of.....Ex-situ Conservation
- 46: Areas of great genetic diversity are protected from human interference..... Gene sanctuary
.....

#Quiz I & #Mid Term MCQ's

47. PGRFA stands for.....plant genetic resource for food and agriculture
48. NCCP stand for.....National cultural collection of Pakistan
49. Conservation of plant genetic resources is necessary for..... Option d
a. food security b. Agrobiodiversity c. Commercial use d. both a & b
50. Peste des petitis ruminants also called..... sheep & goat plaque
51. According to FAO species of molluscus are approx; 85000
52. GSPC was found in..... 1999
53. Which disease was spread by consuming mercury poisoned fish..... MINAMTA disease
54. Plant diversity is urgently and effectively conserved is an objective of..... GSPC
55. Woody plants like trees have__ Genetic diversity than vascular plants like grasses..... Higher
56. Genetic stocks can be divided into ____ groups..... 3
57. Pakistan is the ____ largest producer of Kinow and oranges in world..... 6th
58. Sheep were domesticated by humans around..... 10,000 BC.
59. A _____ can be taken to analyze G cross E interactions..... polygenic approach
60. _____causes advantageous genes to be removed in the following generations.....Genetic drift

61. ____ is an example of daily migration..... Crow
62. ____ be through sexual or asexual reproduction..... Vertical gene transfer
63. ____ is an example of Allopatric speciation..... Darwin finches
64. ____ produced by dart-poison frog..... Toxins
65. Sterculia khasians is endemic tree of the..... Khasi hillss
66. OHSS is..... Ovarian hyper stimulation syndrome
67. Genetic resource is..... first resource
68. Himalayan brown bear..... 150-200
69. Cryopreservation in liquid nitrogen..... -196C
70. Special care unit..... Ex-situ
71. Closely related individuals..... inbreeding
72. Microorganisms.....diverse
73. National strategy of PGRFA may help..... Country
74. Crustaceans migrate for..... breeding/mating
75. Phenyl ketonuria caused by..... Mutations
76. Genetic drift effects on storage in..... small population
77. More genetic variations in population.....Natural selection
78. Coral reefs are home of marine animals.....25%
79. Earliest modern protected areas..... yellow stone n.p
- 80.Ccalypha rubrinervis belongs to family..... Euphorbiaceae

[#Bt401](#) at 11am (4/9/2018)

[#Objective:-](#)

Mcq's

1. (Obsolete varieties) are varieties developed since the advent of scientific agriculture in the late nineteenth century.
2. (Ex-situ) conservative can be used to reintroduce species that left an area.
3. (Acacia nilotica) is easily regenerated from seed.
4. Anthropogenic impact is apparent in the (coastal zone).
5. mild breeding is type of (Inbreeding)
6. The problem in strict natural Nature reserve is Option d
(a) Disease (b) climate change (c) air (d) both b & c
7. (National monument) are small sites they focus on more prominent natural features.
8. IBPGR stands for (International Board of Plant Genetic Resources)
9. In (2014) the food and agriculture organisation of UN published the first State of Worlds FGR.
10. The Wild Birds abd Animals Protection Act (1912)
11. The pakistan Animal Quarantine Ordinance (1979)

Short Question:-

1. Role of crypcryptant in cryopreservation (3 max)

[#OtherRepeated](#)

MCQ BT401 Final Term and Midterm spring 2018

Final Term MCQs

Q1: Which of the following can be used to protect the individual animal in controlled environment?

Ex Situ Conservation

Q2: Total number of snow leopard found in Pakistan

300

Q4: Which of the following is a result of reproductive isolation?

Reproductive Isolation

Q5: Bees are responsible for the pollinating more than ----- species of plants

250,000

Q6: Most apparent problem in strict nature reserve is with

Climate and Air Pollution

Q8: ----- are generally quite small protected areas and often have high visitor value

National- Monument Feature

Q9: WCPA stands for

World Commission on Protected Areas

Q10: Which of the following category of protected areas act as a buffer around a core of one or more protected areas?

Category V, Protected Landscape/seascape

Q11: Category VI is not designed to accommodate

Large-scale industrial harvest

Q12: National Park spread in an area of

100 sq Km to 500 sq KM

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MCQ BT401 Final Term and Midterm spring 2018

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1970

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Animals

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50 to 200

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Seed Storage

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MCQ BT401 Final Term and Midterm spring 2018

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Q43: Species that are threatened are sometimes characterize by

The population dynamics measure of critical dispensation

Q44: Primary objective of strict nature reserve is to conserve

Regionally • nationally • globally outstanding (All Above)

Q45: Which of the following culturally influenced feature

Cave dwelling

Q46: Earthworm, ants and termite have been described as

Ecosystem engineers

Q47: Tree population is rely on ----- interplaying mechanisms to respond to
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3

Q48: According to FAO there are ----- species of seaweed

13000

Q49: What is the main reason of bottle neck effect in elephant seals

Hunting

MCQ BT401 Final Term and Midterm spring 2018

Q50: ----- encourage the conservation of agro-biodiversity and aquatic biodiversity

Protected landscape/seascape, category V

Q 51: Chashma and Taunsa Barrage declared wild life sanctuary by the
Punjab Govt.

Q52: Main reason of great Indian bustard to become critically endangered
Hunting, Habitat loss, fragmentation (All above)

Q53: ----- is a key factor regulating many of biochemical processes
Temperature

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Herbivorous insects

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MCQs BT401 Midterms Spring 2018

Q1: Generic Resource for food, agriculture and forestry includes
Both wild species and domesticated

Q2: Average life of modern varieties is
5 to 10 years

Q3: Which one of following is example of in situ?
Both above (National park and biosphere reserve)

Q4: ----- is known as Indian rosewood
Dalbergia Sissoo

Q5: The area under major pulse crop in Pakistan is about
1.5 M hectares

Q6: Sheep were domesticated by human in -----BC

MCQ BT401 Final Term and Midterm spring 2018

10,000

Q7: Example of extinct species is

White Rhino

Q8: Genetic Diversity is the variation of individual in

Genes

Q9: Indus River Dolphin is

Mammal

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All of the above (Local Climate change, Availability of food, seasons of the year)

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Q16: Which one of the following is example of artificial barrier?

All of the above (The Great Wall of China, Dams, barrages)

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Horn of Rhino

Q18: The Tecopa pupfish was native to

MCQ BT401 Final Term and Midterm spring 2018

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Q19: Which one of the following plant material can be selected?

All of the above (Meristem, embryo, ovule)

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MCQ BT401 Final Term and Midterm spring 2018

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MCQ BT401 Final Term and Midterm spring 2018

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Q60: There are only ----- snow leopard left in Pakistan

200

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Q62: Land Race is ----- of domesticated plant specie

Local variety

Q63: GSPC program founded in

1999

Q64: Estimated total remaining snow leopard worldwide is

7000 to 10000

Q65: According to the World Database on Protected Areas (WDPA) there are over-----

protected areas around the world

210,000

Q66: How many categories of vertical gene transfer?

2

Q67: Breeding type that is not related to half sister half brother and cousins

Line breeding

Q68: The strongest effect of genetic drift on-----

Small population

Q69: Themes of genetic resources

MCQ BT401 Final Term and Midterm spring 2018

2

Q70: Gene flow blocked by physical barriers

Allopatric

Q71: Germplasm of a crop may be defined as the sum total of all hereditary material. i.e. -----

All the Alleles of genes

Q72: Microorganisms produce

Antibiotics

Q73: Genetic stock can be divided into

a) Cytological stock b) mutant stock c) germplasm set

All

Q74: Area of genetic diversity is protected from human interfere

Gene sanctuary

Q75: Brain is more developed in

Cat

MCQ BT401 Final Term and Midterm spring 2018

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7000 to 10000

Q65: According to the World Database on Protected Areas (WDPA) there are over-----
protected areas around the world

210,000

Q66: How many categories of vertical gene transfer?

2

Q67: Breeding type that is not related to half sister half brother and cousins

Line breeding

Q68: The strongest effect of genetic drift on-----

Small population

Q69: Themes of genetic resources

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2

Q70: Gene flow blocked by physical barriers

Allopatric

Q71: Germplasm of a crop may be defined as the sum total of all hereditary material. i.e. -----

All the Alleles of genes

Q72: Microorganisms produce

Antibiotics

Q73: Genetic stock can be divided into

a) Cytological stock b) mutant stock c) germplasm set

All

Q74: Area of genetic diversity is protected from human interfere

Gene sanctuary

Q75: Brain is more developed in

Cat

BT401 Current Final Term spring 2018 Solved Paper

Q1: What is Botanical Garden? 2

It is a garden dedicated to the collection, cultivation and display of a wide range of plants labeled with their botanical names. It may contain specialist plant collections such as tropical plants, or other species of plants.

Q2: Define Forest Genetic Resource? 2

Forest genetic resources (FGR) are the heritable materials maintained within and among tree and other woody plant species that are of actual or potential economic, environmental, scientific or societal value.

Q3: Give different Variations of Gene variations? 3

Genetic refers to variation of genetic ☐ DNA ☐ origin, and variation of genes at different levels: variation between individual trees within populations. The largest variation is between species, and loss of whole species is therefore also the most dramatic loss of future options.

Q4: Define Strict Nature Reserve 3

Protected areas that are strictly set aside to protect biodiversity where human visitation, use and impacts are strictly controlled to ensure protection of the conservation values.

Q5: Define Seed Storage 3

In a genebank, seed storage is the preservation of seeds under controlled environmental conditions which will prolong the viability of the seeds for long periods. Two types of seed stores are used for collections of genetic resources, those holding base collections and those holding active collections. The temperature, relative humidities, seed moisture contents, containers and distribution arrangements vary between these stores.

Q6: Write Features of Cartagena Protocol 5

The Bio safety (Protocol to CBD) deals with; • Safe handling, • Storage • Trans-boundary
n May 2009

Q7: What is Cold storage discuss?

Germplasm of some plants (in the form of shoot tips, nodal or meristem explant culture) are stored at low and non-freezing temperature (1-9°C). At low temperature, growth of plant material is slow down but not completely stopped as in cryo-preservation. In cold storage there is no risk of cold injuries.

Q8: Write the impacts of Climate change on FGR also discuss the effect of climate change on organisms which associated with FGR. 10

Impacts of Climate change on FRG

Climate change may also result in high variability in temperature and precipitation, with an increase in incidence of extreme events, such as flooding, late frosts and intensive summer droughts, amongst other events. In some areas, such as the Mediterranean and the Neo-tropics, an increase in seasonality is also expected. Under such conditions, natural selection may not result in efficient adaptation because selection pressures are multi-directional, involving traits

that may be inversely correlated at the gene level. The standing genetic variation in populations may then not be large enough to create the rare new genotypic combinations that are required. Ecosystems affected by abrupt change may sustain rapid and widespread transformation as ecological tipping points are exceeded. Given the pivotal role of trees in ecosystem function, abrupt climate change impacts on them may thus have profound consequences for forests as a whole. Irreversible loss of ecosystem integrity and function may follow, with replacement by new non-endemic ecosystems.

Effect of climate change on Organisms associated with Trees

In particular, changes in the biology of insect pests and diseases may make ecosystems more susceptible to tree mortality. Because of improved environmental conditions for the pest and reduced tree resistance due to increased stress, pests may react to climate change with range expansions and/or increases in attack severity.

Q9: Briefly explain state of the world's forest genetic resource 3

In 2014, the Food and Agriculture Organization of the United Nations published the first State of the World's Forest Genetic Resources. The publication addressed the conservation, management and sustainable use of forest tree and other woody plant genetic resources of actual and potential value for human well-being in the broad range of management systems.

Q10: What are the three Pillars of Ramsar Convention? 5

The “three pillars” of the convention The Contracting Parties(160)commit to:

- Work towards the wise use of all their wetlands.
- Designate suitable wetlands for the list of Wetlands of International Importance.
- Cooperate internationally on transboundary wetlands, shared wetland systems and shared species

Q11: What is the Extreme weather condition of invertebrate genetic resource? 5

Extreme weather events such as heat waves, droughts and floods – which are predicted to

increase in frequency due to climate change – are often followed by pest outbreaks. Among other contributing factors, these outbreaks can occur because the extreme event eliminates or weakens a pest's natural enemies.

Q12: What type of sources provided by PEPA? 5

Pakistan Environment Protection Act 1997

PEPA provides for; • Protection, • Conservation, • Rehabilitation and • Improvement of the environment;
sustainable development.

Q13: What areas are covered by Cartagena Protocol? 5

□ transboundary movement, transit, handling and use of all living
modified organisms that may have adverse effects on the conservation and sustainable use of
biological diversity, taking into account risks to human health □.
are pharmaceuticals for humans that are addressed by other relevant international agreements or
organizations

Q14: Briefly explain historical background of Ramsar Convention. 3

Ramsar is one of the global inter-governmental environmental agreements. • The treaty was
negotiated in 1960s by countries and NGOs.

To avoid the increasing loss and degradation of wetland habitat for migratory water birds • In a
18 nations meeting it was adopted in the Iranian city of Ramsar On 2nd February 1971 • Came
into force in 21st December 1975

Q15: Discuss purposes of penal code. 2

These punishment are of following types; • Punishment for water pollution • Punishment for
atmospheric pollution • Punishments for general pollution

Q16: What is the location of Cholistan Wildlife sanctuary? 2

Cholistan Wildlife Sanctuary is part of the Cholistan desert in the south eastern portion of the

province of Punjab.

Q17: Mission of Ramsar Convention

The convention's mission is the conservation and wise use of all wetlands through local and national actions and international cooperation, as contribution towards achieving sustainable development throughout the world.

Q18: Which is the second largest national park? 2

Kirthar National Park is the the second largest national park of Pakistan spread over an area of 3000 square kilometres.

Q19: When and which city Cartagena protocol? 2

Cartagena protocol was adopted on June 2001 in Cartagena, Spain

Q20: What is Cholistan protected area name? 2

Cholistan Wildlife Sanctuary is part of the Cholistan desert in the south eastern portion of the province of Punjab

Q21: Write the Objectives of Monument feature 3

Primary

- To protect specific outstanding natural features
- Their associated biodiversity and habitats.

Others

- To provide biodiversity protection in landscapes or seascapes that have otherwise undergone major changes
- To conserve traditional spiritual and cultural values of the site.

Q22: Strategies for Animal Genetic Resource 5

Following strategies should be followed for effective conservation and utilization of AnGR:

1. Formulating the National Livestock and wild-life Breeding Policies.
2. Encouraging the Formation of Breed Associations.

3. Developing Professional Human Resources.

4. Strengthening Research and Development Institutions.

5. Developing Infrastructure for marketing

Q23: Name of three Public Botanical Gardens of Pakistan 3

-e-Jinnah, Lahore

Q24: Name of 3 research based Botanical gardens 3

Quaid-e- rdens (part of Gatwala Wildlife Park),

Q25: Plant Genetic Resource 3

and plant breeders use to improve t

any genetic material of plant origin of actual or potential value for food and agriculture, e.g.

seeds, tubers, mature plants etc.

I not only in crop improvement

includes all our agricultural crops and even some of their wild relatives because they too often have valuable traits.

Q26: Objective of Cartagena 5

and use of living modified organisms resulting from modern biotechnology that may have adverse effects on the conservation and sustainable use of biological diversity, movements.

odified organisms (LMOs)

resulting from modern biotechnology.

Q27: Write a note on conservation of botanical garden. 5

conservation of rare and threatened plants. The conservation of plant diversity is critical for sustainable development and botanic gardens are playing a key role as centers of conservation action. * Botanical gardens can promote diversity. Because they include many species of plant

Q28: Who established Bahawalpur Zoo?

Nawab of former state Bahawalpur, Sir Sadiq Muhammad Khan Abbasi, established Bahawalpur Zoo in 1942

Q29: National strategy for conservation of PGR?

A National Strategy for PGRFA is the blueprint for the management of a country's PGRFA as a continuum of interventions in order to achieve clearly defined time bound goals.

-designed National Strategy needs to be tailored to the particular circumstances and r national, regional and global conservation strategies or initiatives.

Q30: Conservation Status of Hingol national park.

Hingol National Park spread over an area of about 1,650 square km along the Makran Coast, ibexes, an

birds are supported by this park

Q31: In-vivo gene bank

Generally plant seeds, vegetative propagules are used for storage for long time. The whole plants are preserved. This type of conservation strategy is called in vivo gene bank. In this approach, conservation method of storage is used for preservation of plant genetic resources

Q31: Socio-cultural value of national monument feature

Category III areas are likely to hold socio-cultural values as they may have features such as sacred groves, springs, waterfalls, mountains, sea coves etc. of importance to one or more faith groups. These areas are often of significant tourism value and can be managed with the objective of promoting sustainable tourism.

Q32: Different types of Gene bank 2

(i) In vivo Gene Bank:

Generally plant seeds, vegetative propagules are used for storage for long time. The whole plants are preserved. This type of conservation strategy is called in vivo gene bank. In this

approach, conservation method of storage is used for preservation of plant genetic resources

(ii) In vitro Gene Bank:

This approach includes the conservation of genetic resources by nonconventional methods. In this approach explants are grown on medium.

Q33: Limitations of Germplasm Conservation 5

- 1) seeds of many important tree species cannot adequately survive long-term storage;
- (2) seeds of many tropical hardwoods cannot survive even a year of conventional seed storage;
- (3) some genetic damage or change in gene frequencies may occur during seed storage.

Cryogenic storage at its current level of technology offers great promise for long-term storage of true orthodox and sub-orthodox seeds. It has potential for application to recalcitrant species, but much more research and development will be required. Some advantages of cryogenic storage are: (1) it can apparently extend storage life of true-orthodox species far beyond that possible in conventional storage;

(2) genetic damage may be much less than what occurs in conventional storage.

Disadvantages include (1)special equipment is required, which is not readily found in forestry facilities; (2) it is costly for large seeds (3) recalcitrant seeds (both classes) cannot be stored cryogenically with current technology.

Q34: Define phenotypic plasticity 2

Phenotypic plasticity is defined as the capacity of a particular genotype to express different phenotypes under different environmental conditions.

Q35: What are habitat species management area 2

IUCN Management Category IV (Habitat/Species Management Area) refers to areas that are managed to protect particular species or habitats. They are defined by IUCN as “protected areas aiming to protect particular species or habitats and management reflect this priority.

To maintain, conserve and restore species and habitats.

Q36: Primary Objective of National Park 2

The primary objective of this category is to protect the natural biodiversity also its ecological structure and the other processes which supports the natural environment. Other objectives include managing the area in such a way that the natural stat perpetuate as possible in its natural form. Ecologically functional and viable populations of native species are maintained, and also to contribute in the revenue generation via tourism

Q37: Threats to FGR 5

Forests are under threat, chiefly as a result of human activities, including climate change. On a continent-wide scale, changes in rainfall patterns and temperature mean that some tree species will be unable to survive in their current locations. Others may be able to colonize new areas. Changes in climate also make it possible for pests and diseases to invade new areas, destroying the forests there. Overall, we can be sure that the composition and distribution of forests will change. It is because humans – and indeed the whole planet – derive so many benefits from forests and trees that we need to be concerned about how they will adapt to climate change

Q38: Links between botanical garden and climate change 5

Plants can alter the temperature of the Earth's atmosphere. Through the process of photosynthesis, plants use energy from the sun to draw down carbon dioxide from the atmosphere and then use it to create the carbohydrates they need to grow. Since carbon dioxide is one of the most abundant greenhouse gases, the removal of the gas from the atmosphere may temper the warming of our planet as a whole. *transpiration in plants can increase water vapor in the atmosphere, causing more precipitation and cloud cover in an area. The additional cloud cover often reinforces the cooling by blocking sunlight. *Contribute to soil fertility and prevent soil erosion

Q39: Name of labs hosted by Plant genetic resource institute 5

1. germplasm exploration lab

2. seed preservation lab
3. in vitro conservation lab
4. germplasm evaluation lab
5. plant introduction and seed health lab
6. Data management lab

Q40: What is treaty 2

Treaty, a binding formal agreement, contract, or other written instrument that establishes obligations between two or more subjects of international law”

Q41: Primary Objective of Category VI 3

- To protect natural ecosystems
- Use natural resources sustainably, when conservation and sustainable use can be mutually beneficial.

Q42: Explain method of “processing of Sample” in DNA bank 5

DNA preserved in DNA banks will be stored either within cells and extracted upon retrieval from storage or extracted from cells and purified before storage. The quality of the DNA is expressed through yield, purity, molecular weight, amplification efficiency and authenticity of sequences. The quality of DNA extracted from plant specimens is dependent on the condition of the specimen before storage, the storage environment and the duration of storage. Rapid drying of plant samples with silica gel or lyophilisation helps to preserve the DNA.

Q43: How invertebrate genetic resource cope with climate change? 5

Because of the many ecosystem services that they provide, invertebrates have a key role to play in adapting agriculture to the effects of climate change. The extent to which the individual services provided by invertebrates will be enhanced or impeded by climate change is difficult to predict. However, if invertebrate biodiversity is lost, the capacity of ecosystems to adapt is likely to diminish.

Healthy soils – and healthy, diverse soil invertebrate communities – will be vital to climate change adaptation. For example, earthworms help to maintain soil structure and the availability of water throughout the soil profile. Studies have shown that the presence of these animals can help to alleviate the effects of drought on crop Production. Studies have also revealed the remarkable ability of diverse soil invertebrate communities to restore the structure of degraded soils

The potential for managing soil invertebrates to enhance their beneficial roles has been little explored. Few if any deliberate attempts have been made to introduce soil invertebrates into new countries or ecosystems. Given the potential for such species to become invasive, it is inadvisable to attempt any such introductions until soil ecology is much better understood than it is today. However, every effort should be made to avoid agricultural practices that disrupt resident soil invertebrate communities and the services they provide.

It is likely that some pests, as they move into new areas in response to climate change, will at least temporarily □escape□ from their natural enemies. This is likely to increase demand for classical biological control agents, especially in places where the newly established pest population is separated from its original home by a physical barrier such as the sea or a mountain range. For this reason, access to new classical biological control agents is likely to be particularly important for island countries

Q44: Describe the Features Pakistan Terrestrial and Maritime zone act 1976 5
management of living resources in Pakistan's Exclusive Economic Zone (EEZ)
Government of Pakistan.

Q45: List of Ramsar Sites in Punjab

- Uchhali Complex • Taunsa Barrage • Chashma Barrage

Q46: Two Principles of Successful Quarantine 5

3. When germplasm must be planted and grown for the purposes of quarantine testing, it should

be done in an area geographically and ecologically separated from the major growing areas for that crop, to prevent the establishment of crop-specific pests or pathogens.

4. When germplasm is endangered or the need for particular accessions is particularly urgent, some discretion should be possible on the part of quarantine officials in allowing exceptions for controlled entry, despite existing regulations to the contrary.

Q47: What happen if biodiversity of invertebrates is lost 2

If invertebrate biodiversity is lost, the capacity of ecosystems to adapt is likely to diminish.

Q48: What is EEZ 2

Exclusive Economic Zone

Q49: Give some objective of habitat specie management area

To maintain, conserve and restore species and habitats.

To protect vegetation patterns.

2. To protect fragments of habitats as components of landscape or seascape-scale conservation strategies.

3. To develop public education and appreciation of the species or habitat

Q49: What is storage strategy in DNA Bank

Determining what to store and for how long is an important consideration, used to determine sample size, capacity of the DNA bank, preparation of samples and documentation. Longterm needs and expected volume and number of samples to be stored will determine organization and repository design.

Q50: Historical Background of CBD

CBD is about the conservation and wise use of different biological resources (plants and
dopted in 1992 at Rio De Janeiro, Brazil and entered force on January, 1993,
tan

ratified it on 26th July 1994.

Q51: Why does Forest Genetic Resource matters?

The high levels of genetic variation that are present within many tree species can be beneficially developed and used by foresters and tree growers. Whereas agricultural crop breeders and farmers often substantially modify the growing environment to suit a specific crop species or variety, tree growers commonly identify species and provenances which can provide some improved levels of the goods and services required even without intensive selection and improvement, or intense management requirements, or major modification of the external environment

Q52: Tissue culture conservation advantages

Source of disease-free material. It is most appropriate for rapid multiplication purposes, dissemination and active collections.